# Course – TFC

Course Instructor

Dr. Umadevi V

Department of CSE, BMSCE

Webpage:https://sites.google.com/site/drvumadevi/

# Unit-5

**Problems That Computers Cannot Solve**

The Turing Machine, Programming Techniques for Turing Machines, Extensions to the Basic Turing Machine, Restricted Turing Machines, Turing Machines and Computers, Definition of Post's Correspondence Problem, A Language That Is Not Recursively Enumerable, An Undecidable Problem That is RE

Other Undecidable Problems

# Problems That Computers Cannot Solve

- Specific Problems we cannot solve using a computer are called “Undecidable”

# Example

| Program to print "Hello World" | Fermat's last theorem expressed as a hello world program |
|---|---|

```c
main()
{
    printf("hello, world\n");
}
```

```c
int exp(int i, n)
/* computes i to the power n */
{
    int ans, j;
    ans = 1;
    for (j=1; j<=n; j++) ans *= i;
    return(ans);
}

main ()
{
    int n, total, x, y, z;
    scanf("%d", &n);
    total = 3;
    while (1) {
        for (x=1; x<=total-2; x++)
            for (y=1; y<=total-x-1; y++) {
                z = total - x - y;
                if (exp(x,n) + exp(y,n) == exp(z,n))
                    printf("hello, world\n");
            }
        total++;
    }
}
```

Hello World Problem: Determine whether a given C program, with given input, prints "Hello World"

Note: In number theory, **Fermat's Last Theorem** states that no three positive integers a, b, and c satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than 2.

# Example

The program (Fermat) takes an input $n$ and looks for positive integer solutions to equation

$$x^n + y^n = z^n$$

If the program finds a solution, it prints `hello, world`

If it never finds integer $x, y, z$ to satisfy the equation, then it continues searching **forever**, and never prints `hello, world`

If the value of $n$ is $2$, then it will find combinations of integers and thus:

For input $n = 2$ the program prints `hello, world`

For any integer $n > 2$, the program will never find a triple of positive integers to satisfy $x^n + y^n = z^n$

```
int exp(int i, n)
/* computes i to the power n */
{
    int ans, j;
    ans = 1;
    for (j=1; j<=n; j++) ans *= i;
    return(ans);
}

main ()
{
    int n, total, x, y, z;
    scanf("%d", &n);
    total = 3;
    while (1) {
        for (x=1; x<=total-2; x++)
            for (y=1; y<=total-x-1; y++) {
                z = total - x - y;
                if (exp(x,n) + exp(y,n) == exp(z,n))
                    printf("hello, world\n");
            }
        total++;
    }
}
```

# Our hello-world problem

- Is there a program H that could examine any program P and any input I for P, and tell whether P, run with I as its input, would print *hello, world*?
- The answer is: undecidable!
  - That is, there exists no such program H.
  - We can prove this by contradiction next.

# Hypothetical "Hello, World" Tester

We want to prove that no program **H**, called hypothetical "Hello, World" tester, as mentioned previously exists by contradiction using the following steps.

**Step 1:** assume H exists in a form as shown in Fig. 8.1

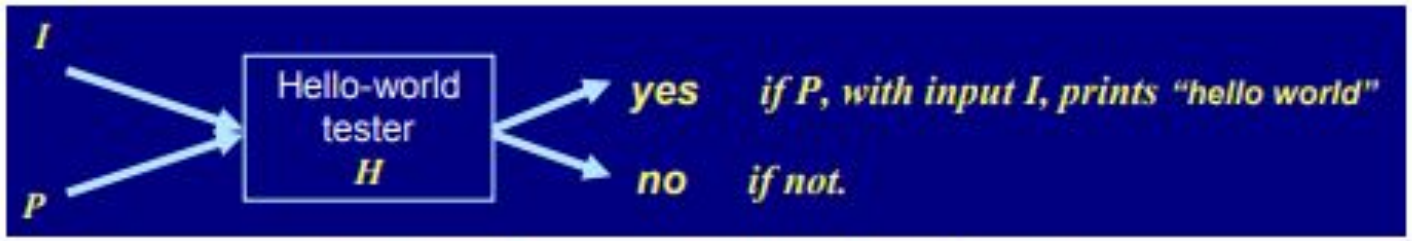



Fig. 8.1 A hypothetical "Hello, World" tester.

**Step 2:** Transform H into another form H2 in a simple way which can be done by C programs.

**Step 3:** prove that H2 does not exist and so that H does not exist, either

# Hypothetical "Hello, World" Tester (Contd…)

Implementation of Step 2 in previous slide

(1) Transform H to H1 in a way as illustrated by Fig. 8.2

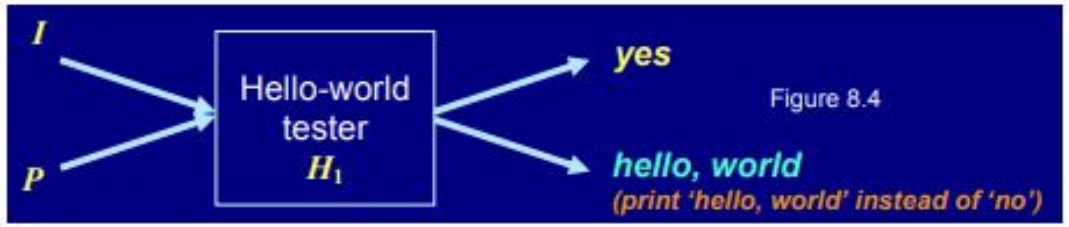



Fig. 8.2 A transformed "hello-world tester" $H_1$.

(2) Transform H1 to H2 in a way as illustrated by Fig. 8.3

Fig. 8.2 A second transformed "hello-world tester" $H_2$.

The function of H2 constructed in **Step 2** is given any program P as input,

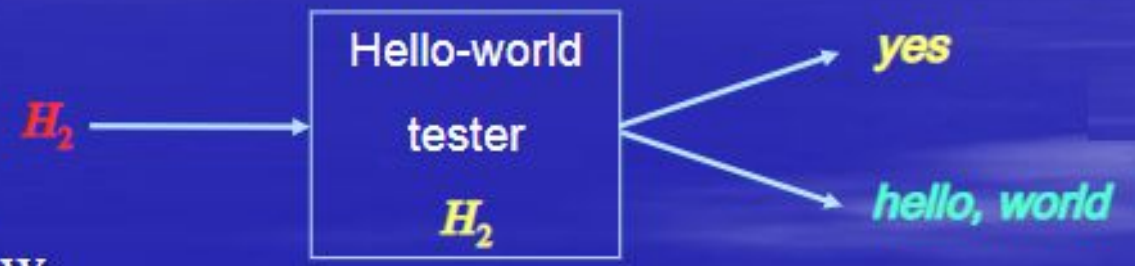
if P prints ***hello, world*** as first output, then H2 makes output **yes**;

if P **does not prints hello, world** as first output, then H2 prints **hello, world**

# Hypothetical "Hello, World" Tester (Contd…)

Implementation of **Step 3** above (proving H2 does not exist) ---

- Prove $H_2$ does not exist by contradiction as follows (cont'd) –




- Now,

(1) if the box $H_2$, given itself as input, makes output *yes*, then it means that

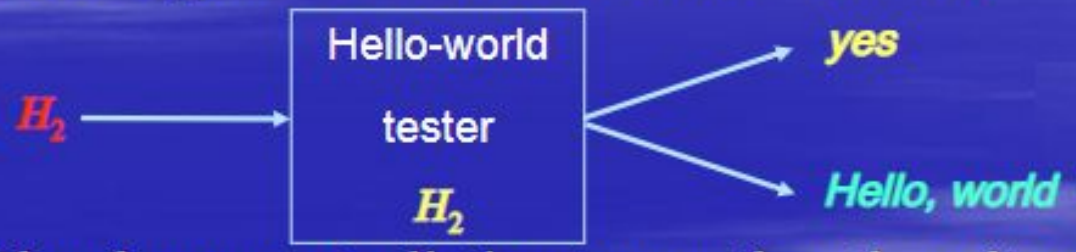
*the input $H_2$, given itself as input, prints* **hello world** *as first output.*

But this is contradictory because we just suppose that $H_2$, given itself as input, makes output *yes*.

# Hypothetical "Hello, World" Tester (Contd…)

Implementation of **Step 3** above (proving H2 does not exist) ---

- Prove $H_2$ does not exist as follows (cont'd) –




- The above contradiction means the other alternative must be true (since it must be one or the other), that is ---

(2) the box $H_2$, given itself as input, prints ***hello, world*** . This then means that

*such* $H_2$, when taken as input $H_2$ to the box $H_2$(itself), will make the box $H_2$ to make output ***yes***.

Contradiction again because we just say that the box $H_2$, given itself as input, prints ***hello, world*** .

# Reducing One Problem to Another

- Now we have an undecidable problem, which can be used to prove other undecidable problems by a technique of *problem reduction*.
  - That is, if we know $P_1$ is undecidable, then we may *reduce $P_1$ to a new problem $P_2$*, so that we can prove $P_2$ undecidable by contradiction in the following way
    - *If $P_2$ is decidable, then $P_1$ is decidable.*
    - *But $P_1$ is known undecidable. So, contradiction!*
    - *Consequently, $P_2$ is undecidable.*
- An illustration of the above idea is illustrated in Fig. 8.4.

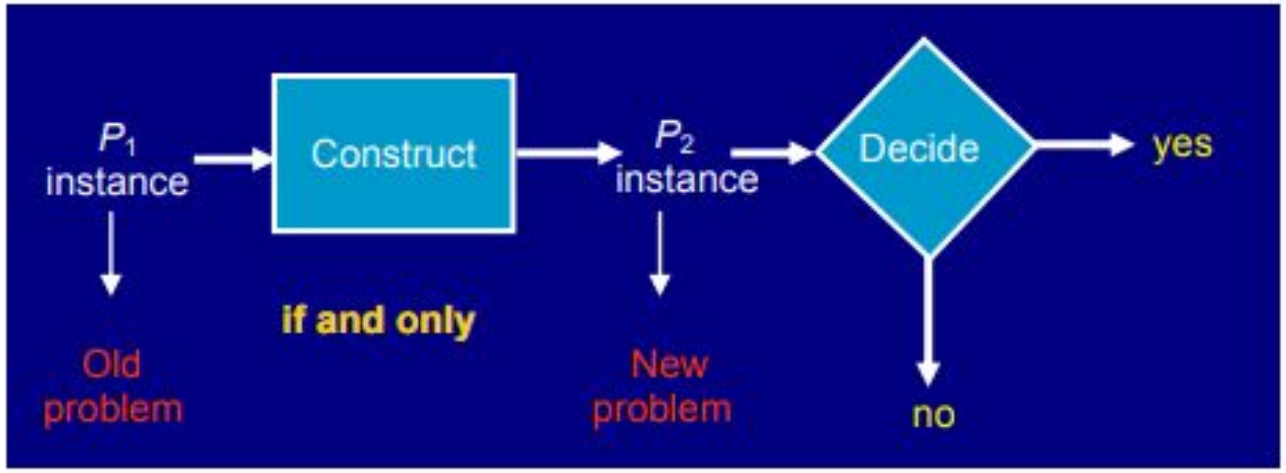



Fig. 8.4 An illustration of reducing one problem to another.

# Propose of the theory of undecidable problems

To provide guidance to programmers about what they might or might not be able to accomplish through programming

**Pragmatic impact**:

- **Intractable problems:**
  - Decidable problems.
  - Require large amount of time to solve them.
  - Contrary to undecidable problems, which are usually rarely attempted in practice, the intractable problems are *faced everyday*.
  - Yield to small modifications in the requirements or to heuristc solutions.

- We need tools that allow us to prove everyday questions undecidable or intractable
- We need to rebuild our theory of undecidability, based not on C programs, but based on a very simple model of computer called **Turing Machine**

# The Turing Machine ™

Informally...

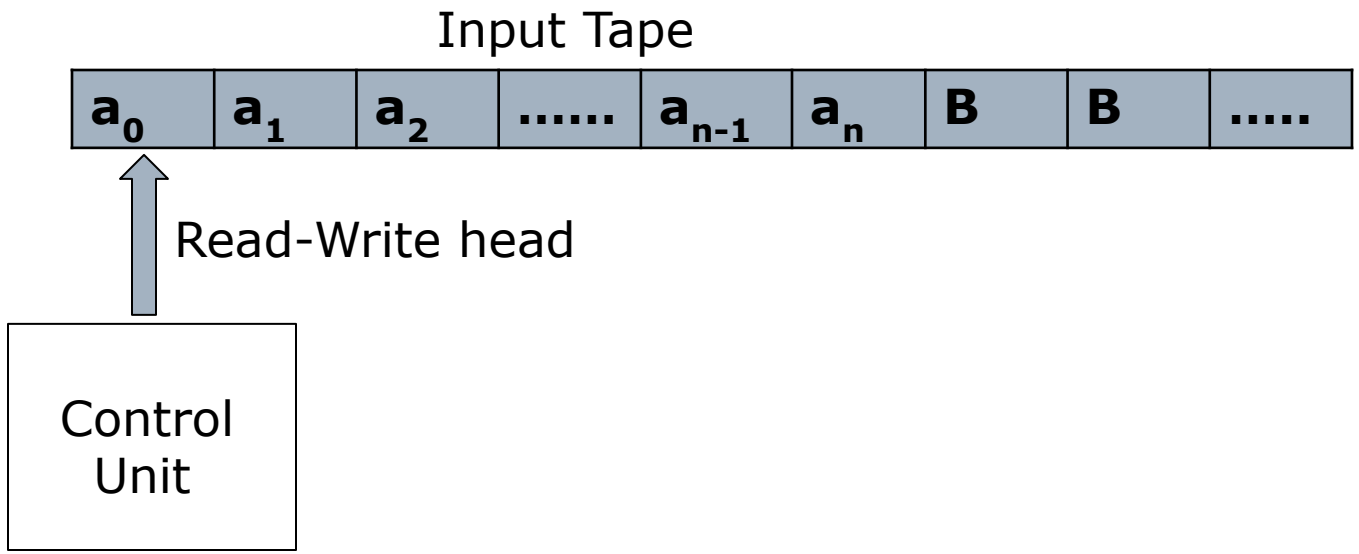
- TM is essentially a **finite automaton** that has a single **tape of infinite length** on which it may read and write data.
- Advantage of TM over programs as representation of what can be computed: TM is sufficiently simple that we can represent its configuration precisely, using a simple notation much like ID's of a PDA.

History

- 1936: Alan Turing proposed the TM as a model of *any possible computation*.
- This model is computer-like, rather than program-like, even though true electronic or electromechanical computers were several years in the future.
- All the serious proposals for a model of computation have the same power; *i.e.*, they compute the same functions and recognize the same languages.

# Turing Machine (TM)

# Turing machine model



# Formal Definition of Turning Machine

$P=(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

- Q is set of **states**
- $\Sigma$ is set of **input alphabet**
- $\Gamma$ is set of **tape symbols**
- $q_0 \in Q$ , is the start state
- B is special symbol indicating blank character
- F is the set of final states
- $\delta$ is a transition function which maps $Q \times \Gamma$ to $Q \times \Gamma \times \{L, R\}$

Note: L and R indicate Left or Right direction move

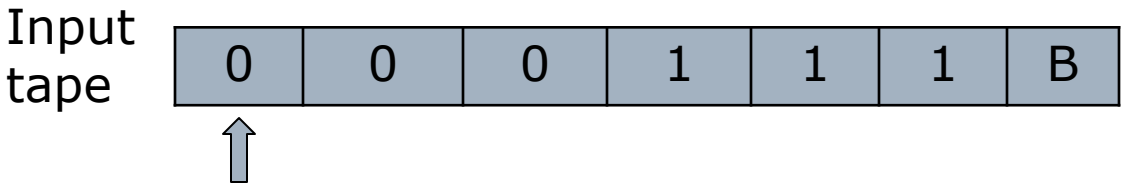
# Question

Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Example:

Valid String: 000111

Invalid String:00111



# Question

Design Turing Machine for L= $\{0^n1^n, n>=1\}$
Example:
Valid String: 000111
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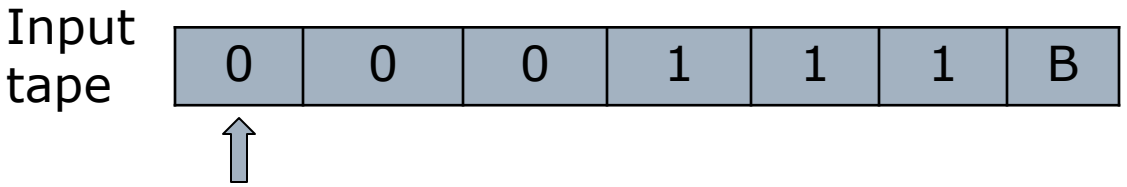


- **Approach used –**
  First replace a 0 from front by X, then keep moving right till you find a 1 and replace this 1 by Y.
  Now keep moving left till you find a X. When you find it, move a right, then
  follow the same procedure as above.
- A condition comes when you find a X immediately followed by a Y. At this point we keep moving right and keep on checking that all 1's have been converted to Y. If not then string is not accepted. If we reach B then string is accepted.

# Rough Slide to Explain "Designing Turing Machine for L= $\{0^n1^n, n>=1\}$"

**Approach used –**

First replace a 0 from front by X, then keep moving right till you find a 1 and replace this 1 by Y.

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| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | B |

| | | | | | | |
|---|---|---|---|---|---|---|
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| | | | | | | |
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|---|---|---|---|---|---|---|
| X | 0 | 0 | Y | 1 | 1 | B |

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| | | | | | | |
|---|---|---|---|---|---|---|
| X | X | 0 | Y | Y | 1 | B |

| | | | | | | |
|---|---|---|---|---|---|---|
| X | X | 0 | Y | Y | 1 | B |

| | | | | | | |
|---|---|---|---|---|---|---|
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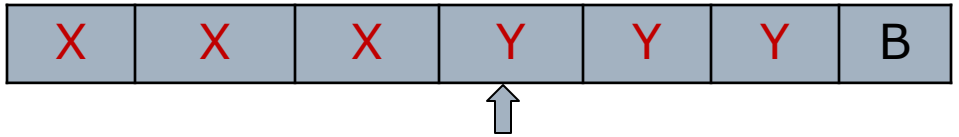

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| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | B |

Invalid String

# Rough Slide to Explain "Designing Turing Machine for L= $\{0^n1^n, n>=1\}$"

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| | | | | | |
|---|---|---|---|---|---|
| X | 0 | Y | 1 | 1 | B |

Invalid String

# Rough to Explain "Designing Turing Machine for L= $\{0^n1^n, n>=1\}$"

**Approach used –**

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Now keep moving left till you find a X. When you find it, move a right, then follow the same procedure as above.

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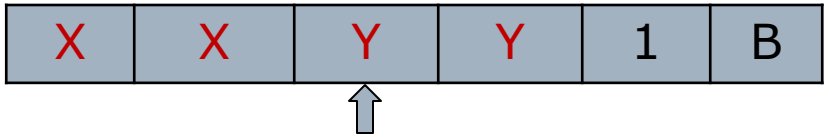


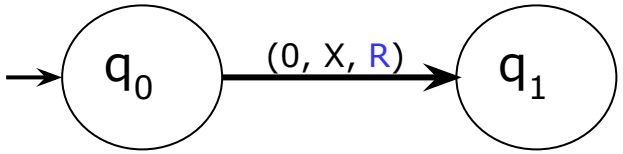

Invalid String

# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

**Logic**

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | B |

**Step-1:** Replace 0 by X and move right,
Go to state q1.



# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

**Logic**

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | B |

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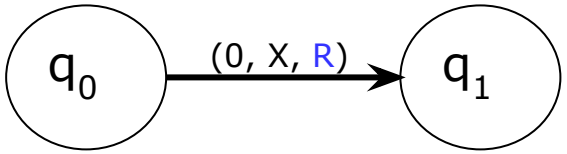
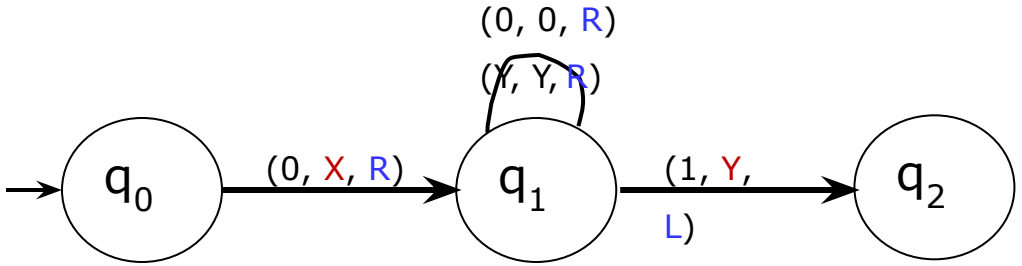



| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | 0 | Y | 1 | 1 | B |

**Step-2:** Replace 0 by 0 and move right, Remain on same state.
Replace Y by Y and move right, Remain on same state.
Replace 1 by Y and move left, go to state q2.

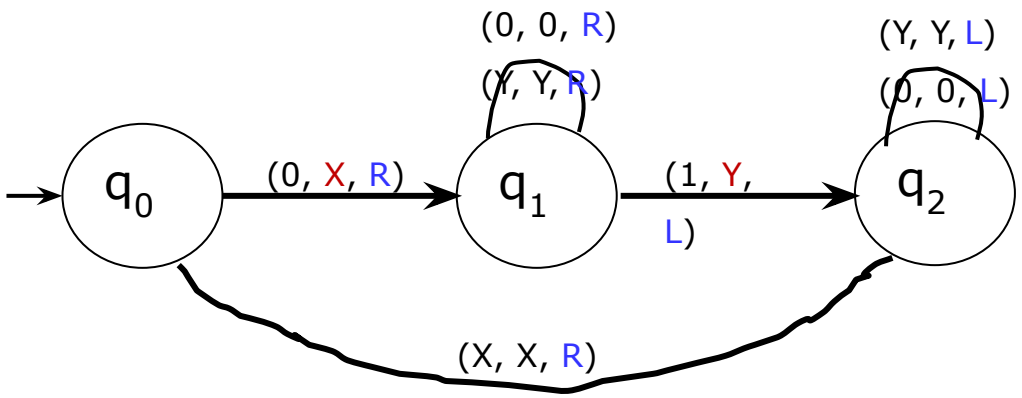


# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Logic

| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | 0 | Y | 1 | 1 | B |

**Step-3:**
Replace 0 by 0 and move left, Remain on same state
Replace Y by Y and move left, Remain on same state
Replace X by X and move right, go to state q0.

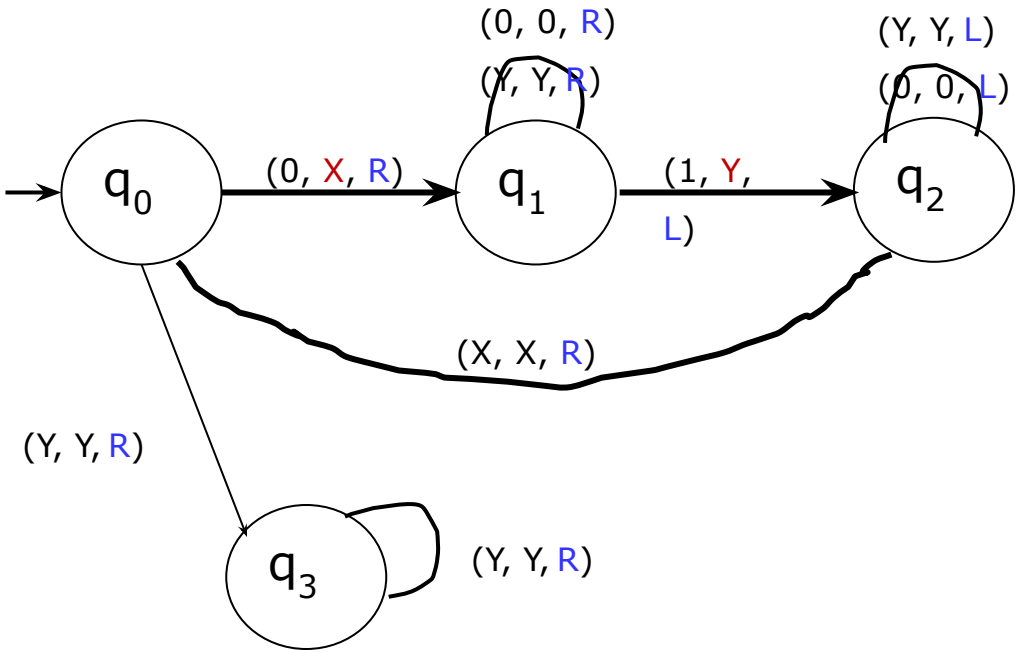


# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Logic

| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | 0 | Y | 1 | 1 | B |

**Step-4:**
If symbol is Y replace it by Y and move right and Go to state q3
Else go to step 1

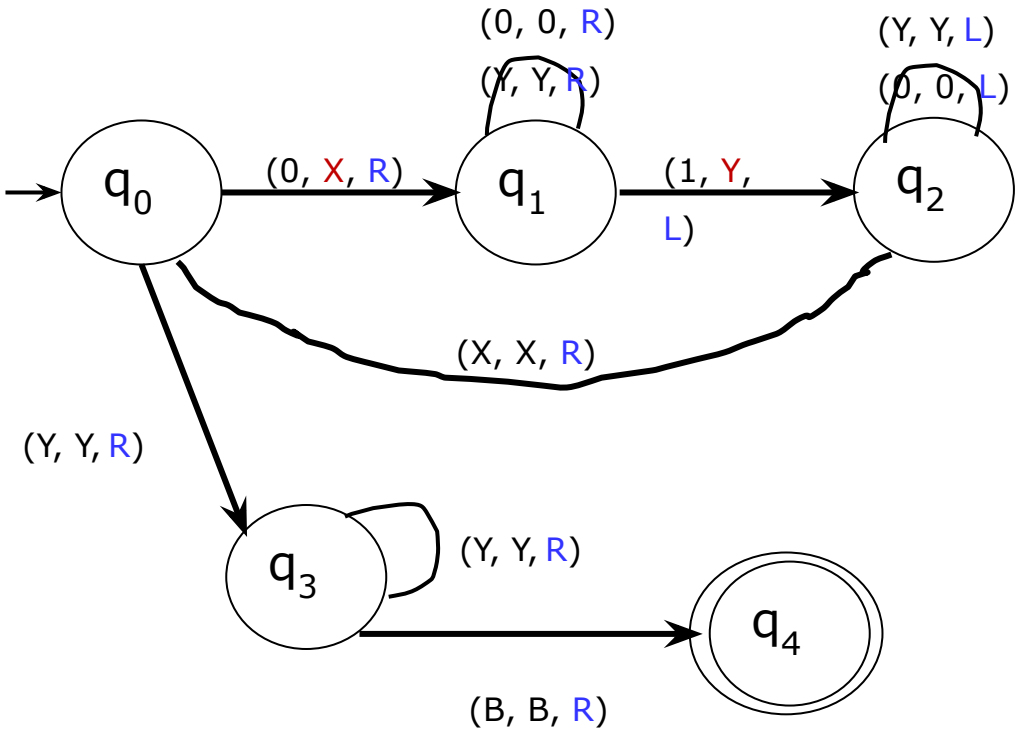


# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Logic

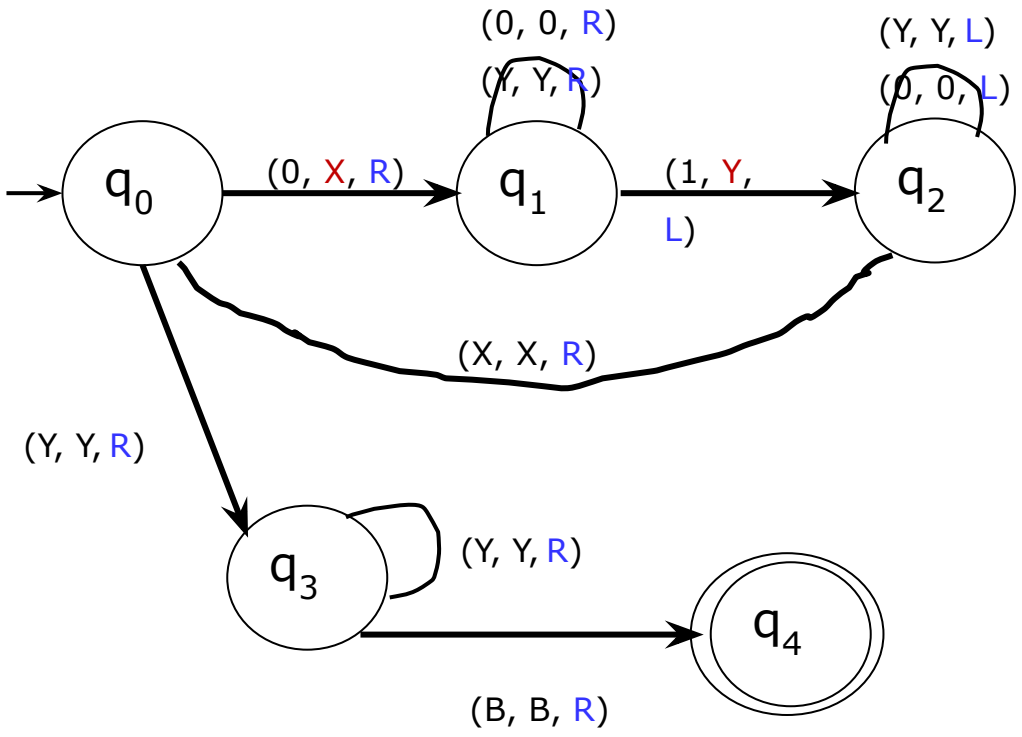
| | | | | | | |
|---|---|---|---|---|---|---|
| X | X | X | Y | Y | Y | B |

**Step-5:**
Replace Y by Y and move right, remain on same state
If symbol is B replace it by B and move right
STRING IS ACCEPTED, GO TO FINAL STATE q4



# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Transition diagram



# Design Turing Machine for L= $\{0^n1^n, n>=1\}$

Formal Definition

$M=(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

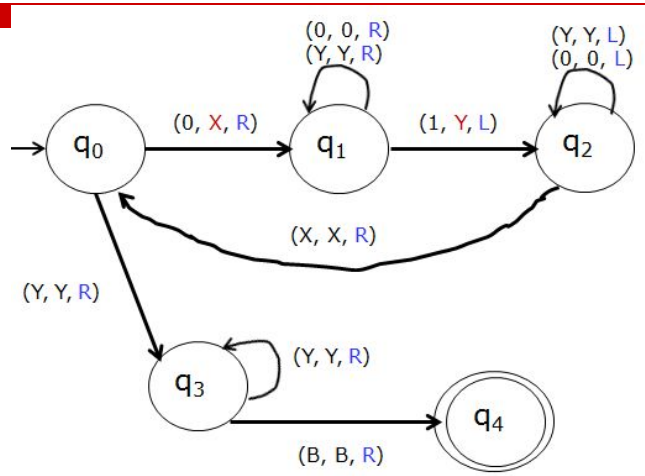
$\Sigma = \{0, 1\}$

$\Gamma = \{0, 1, X, Y, B\}$

$q_0 = q_0$

$B = B$

$F = \{q_4\}$




Transition Diagram

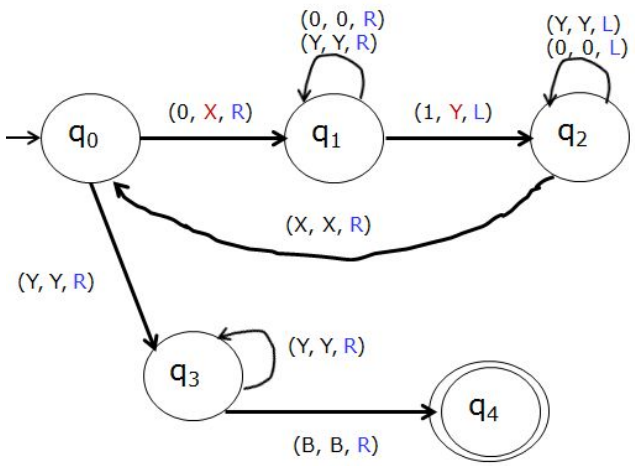
δ Transition table

| State | Input tape Symbols | | | | |
|---|---|---|---|---|---|
| | 0 | 1 | X | Y | B |
| $q_0$ | $(q_1, X, R)$ | --- | --- | $(q_3, Y, R)$ | -- |
| $q_1$ | $(q_1, 0, R)$ | $(q_2, Y, L)$ | --- | $(q_1, Y, R)$ | --- |
| $q_2$ | $(q_2, 0, L)$ | --- | $(q_0, X,R)$ | $(q_2, Y,L)$ | --- |
| $q_3$ | --- | --- | --- | $(q_3, Y, R)$ | $(q_4,B,R)$ |
| $q_4$ | --- | --- | --- | --- | --- |

# TM for L= $\{0^n1^n, n>=1\}$ (Contd….)
## Rough Slide

Instantaneous description for the string 0011

$q_0$0011B



# TM for L= $\{0^n1^n, n>=1\}$ (Contd….)

Instantaneous description for the string 0011

$$
\begin{aligned}
q_0 0011B &\vdash Xq_1 011B \\
&\vdash X0q_1 11B \\
&\vdash Xq_2 0Y1B \\
&\vdash q_2 X0Y1B \\
&\vdash Xq_0 0Y1B \\
&\vdash XXq_1 Y1B \\
&\vdash XXYq_1 1B \\
&\vdash XXq_2 YYB \\
&\vdash Xq_2 XYYB \\
&\vdash XXq_0 YYB \\
&\vdash XXYq_3 YB \\
&\vdash XXYYq_3 B \\
&\vdash XXYYBq_4
\end{aligned}
$$

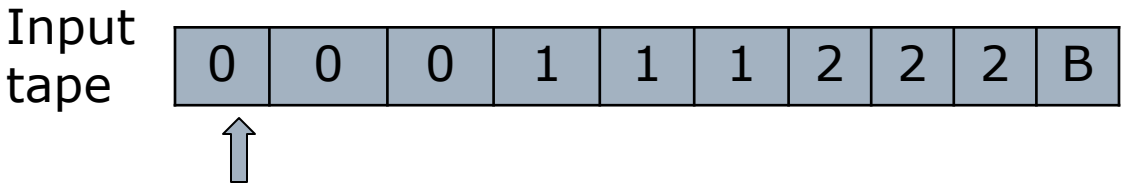
# Question

Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

Example:

Valid String: 000111222

Invalid String:00111222




**Assumption:** We will replace 0 by X, 1 by Y and 2 by Z

**Approach used –**

- First replace a 0 from front by X, then keep moving right till you find a 1 and replace this 1 by Y. Again, keep moving right till you find a 2, replace it by Z and move left. Now keep moving left till you find a X. When you find it, move a right, then follow the same procedure as above.

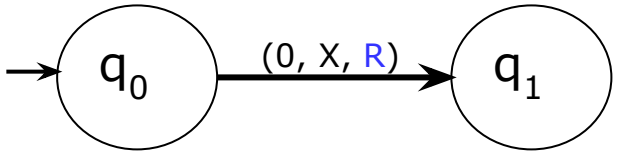
- A condition comes when you find a X immediately followed by a Y. At this point we keep moving right and keep on checking that all 1's and 2's have been converted to Y and Z. If not then string is not accepted. If we reach **B** and final state then string is accepted.

# Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

**Logic**

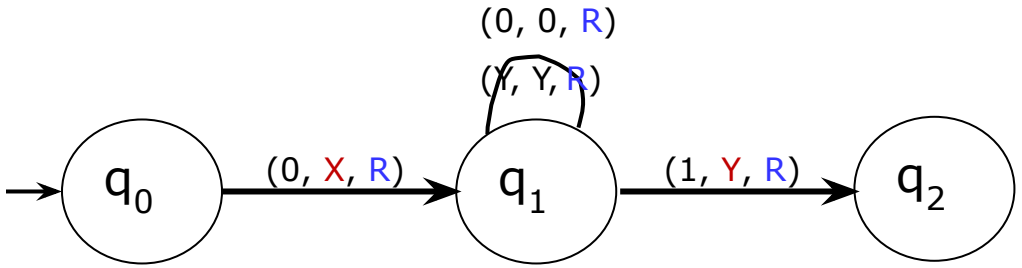
| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 2 | 2 | B |

**Step-1:** Replace 0 by X and move right, Go to state q1.




| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | Y | 1 | 2 | 2 | B |

**Step-2:** Replace 0 by 0 and move right, Remain on same state
Replace Y by Y and move right, Remain on same state
Replace 1 by Y and move right, go to state q2.

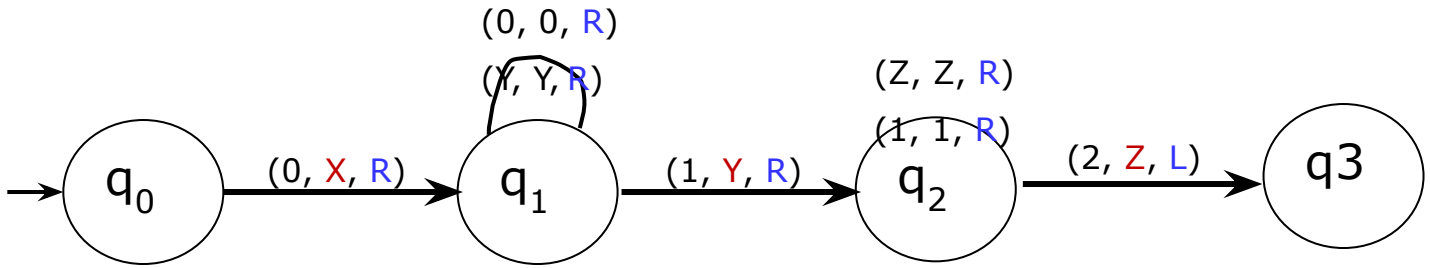


# Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

**Logic**

| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | Y | 1 | 2 | 2 | B |

**Step-3:**
Replace 1 by 1 and move right, Remain on same state
Replace Z by Z and move right, Remain on same state
Replace 2 by Z and move left, go to state q3.

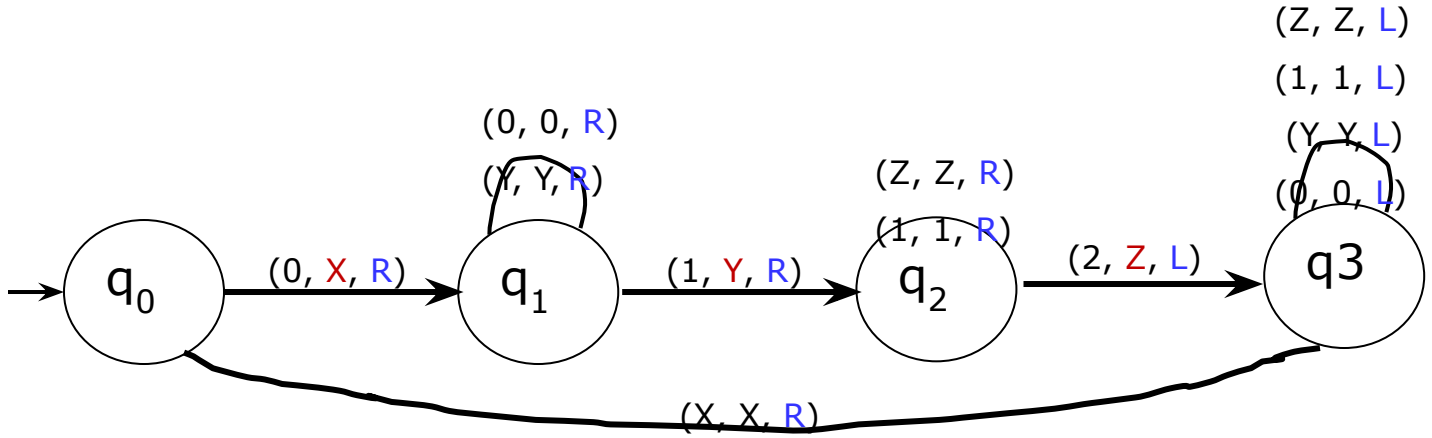


# Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

**Logic**

| | | | | | | |
|---|---|---|---|---|---|---|
| X | 0 | Y | 1 | Z | 2 | B |

**Step-4:**
Replace Z by Z and move left, Remain on same state
Replace 1 by 1 and move left, Remain on same state
Replace Y by Y and move left, Remain on same state
Replace 0 by 0 and move left, Remain on same state
Replace X by X and move right, go to state q0.

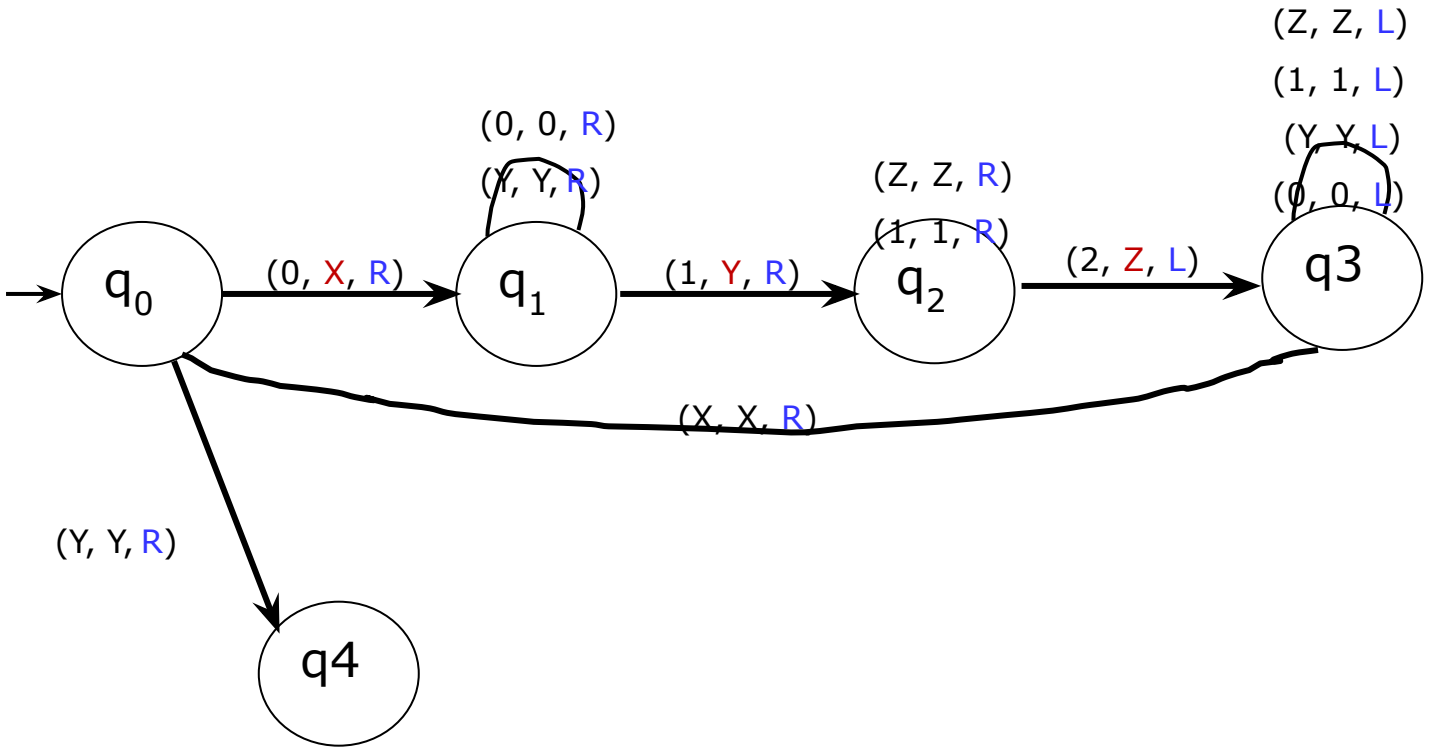


# Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

**Logic**

| | | | | | | |
|---|---|---|---|---|---|---|
| X | X | Y | Y | Z | Z | B |

**Step-5:**
If symbol is Y replace it by Y and move right and Go to state q4
Else go to **Step-1**

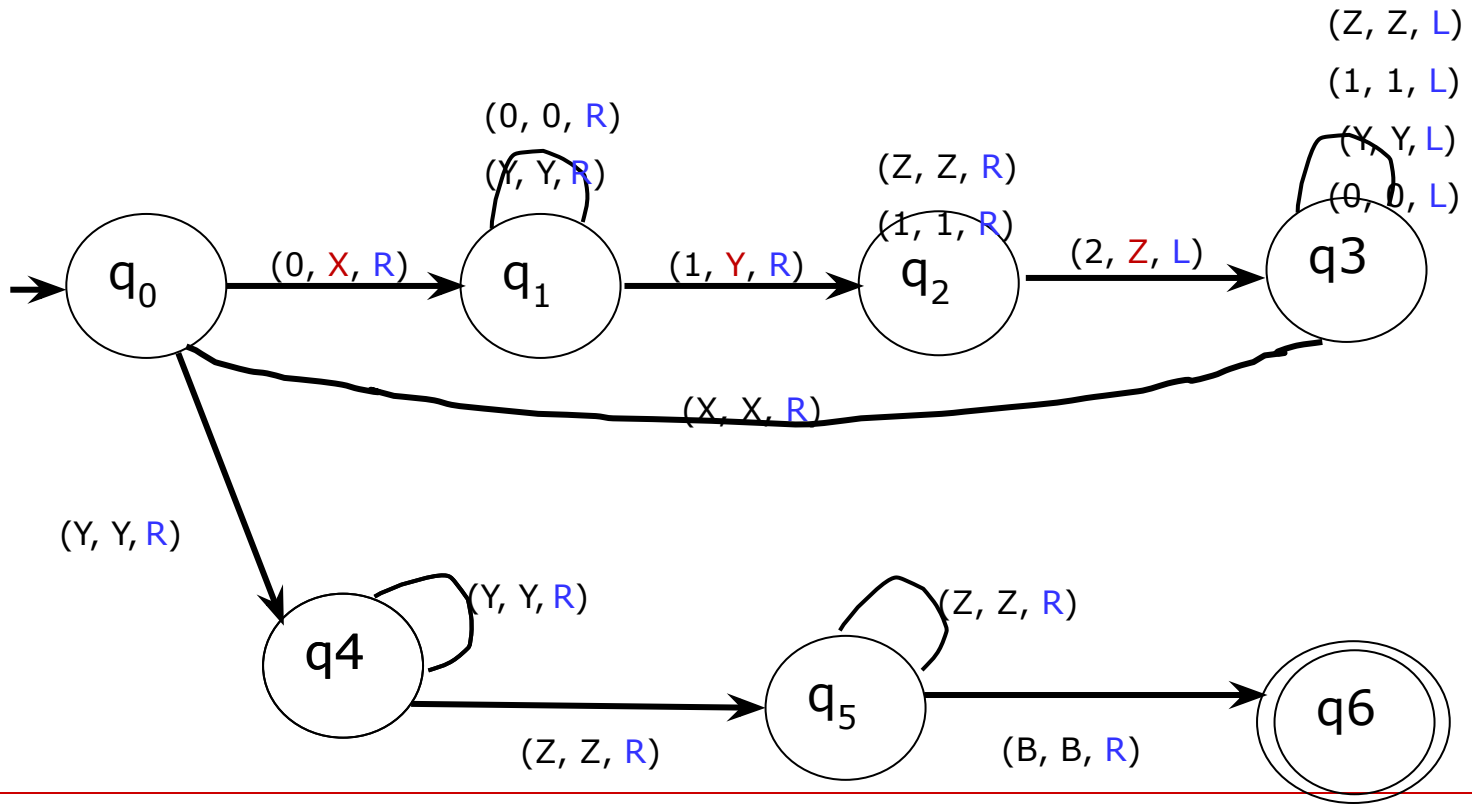


# Design Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

## Logic

| | | | | | | |
|---|---|---|---|---|---|---|
| X | X | Y | Y | Z | Z | B |

**Step-6:**
Replace Y by Y and move right, Remain on same state
If the symbol is Z, replace it by Z and move right and go to the state q5
**Step-7:**
Replace Z by Z and move right, Remain on same state
If symbol is B replace it by B and move right, STRING IS ACCEPTED, GO TO FINAL STATE q6



# Complete design of Turing Machine for L= $\{0^n1^n2^n, n>=1\}$

Formal Definition

$M=(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

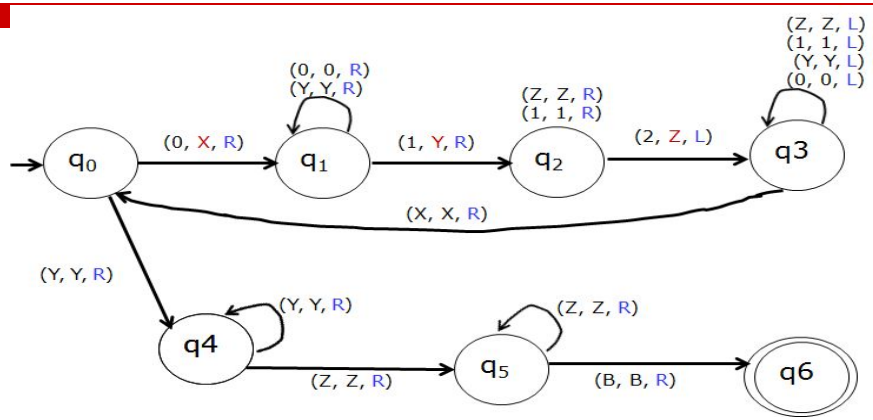
$\Sigma = \{0, 1, 2\}$

$\Gamma = \{0, 1, 2, X, Y, Z, B\}$

$q_0 = q_0$

$B = B$

$F = \{q_6\}$




Transition Diagram

δ Transition table

| State | Input tape Symbols | | | | | | |
|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | X | Y | Z | B |
| $q_0$ | $(q_1, X, R)$ | --- | | --- | (q4, Y , R) | -- | -- |
| $q_1$ | $(q_1, 0, R)$ | $(q_2, Y, R)$ | | --- | $(q_1, Y, R)$ | --- | --- |
| $q_2$ | --- | $(q_2, 1, R)$ | $(q_3, Z, L)$ | --- | --- | $(q_2, Z,R)$ | --- |
| $q_3$ | $(q_3, 0, L)$ | $(q_3, 1, L)$ | -- | $(q_0,X,R)$ | $(q_3, Y,L)$ | $(q_3, Z,L)$ | --- |
| $q_4$ | --- | --- | --- | --- | $(q_4, Y, R)$ | $(q_5, Z, R)$ | --- |
| $q_5$ | --- | --- | --- | --- | --- | $(q_5, Z, R)$ | $(q_6,B,R)$ |
| $q_6$ | --- | --- | --- | --- | --- | --- | --- |

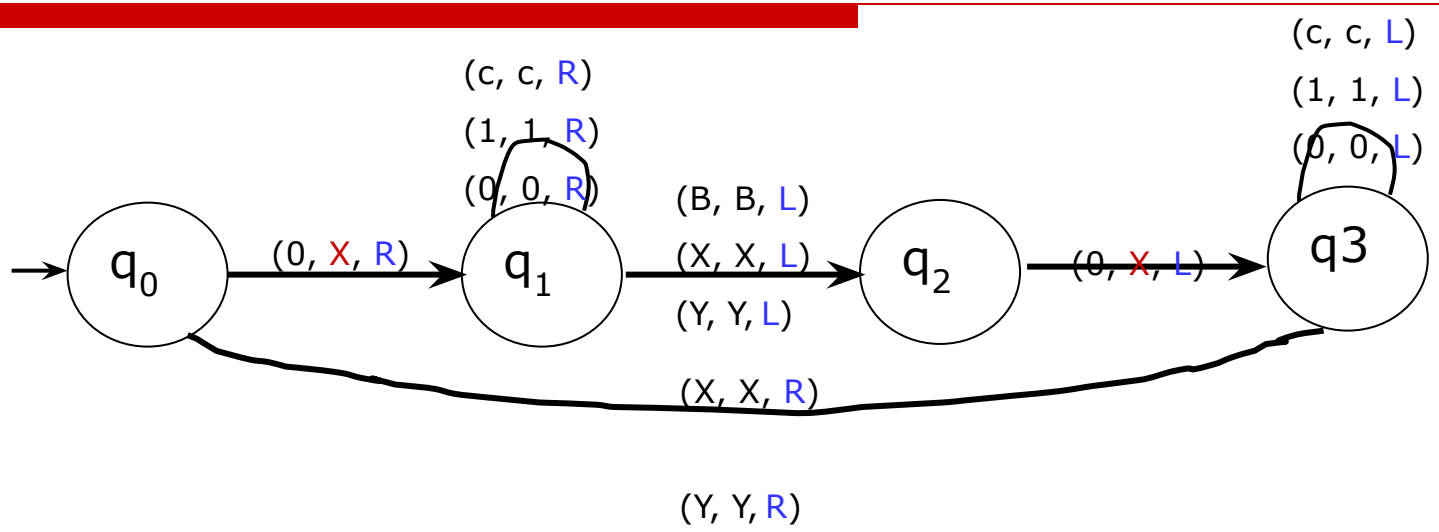
# Question

Design TM for $L=\{wcw^r, w \in (0+1)^*\}$

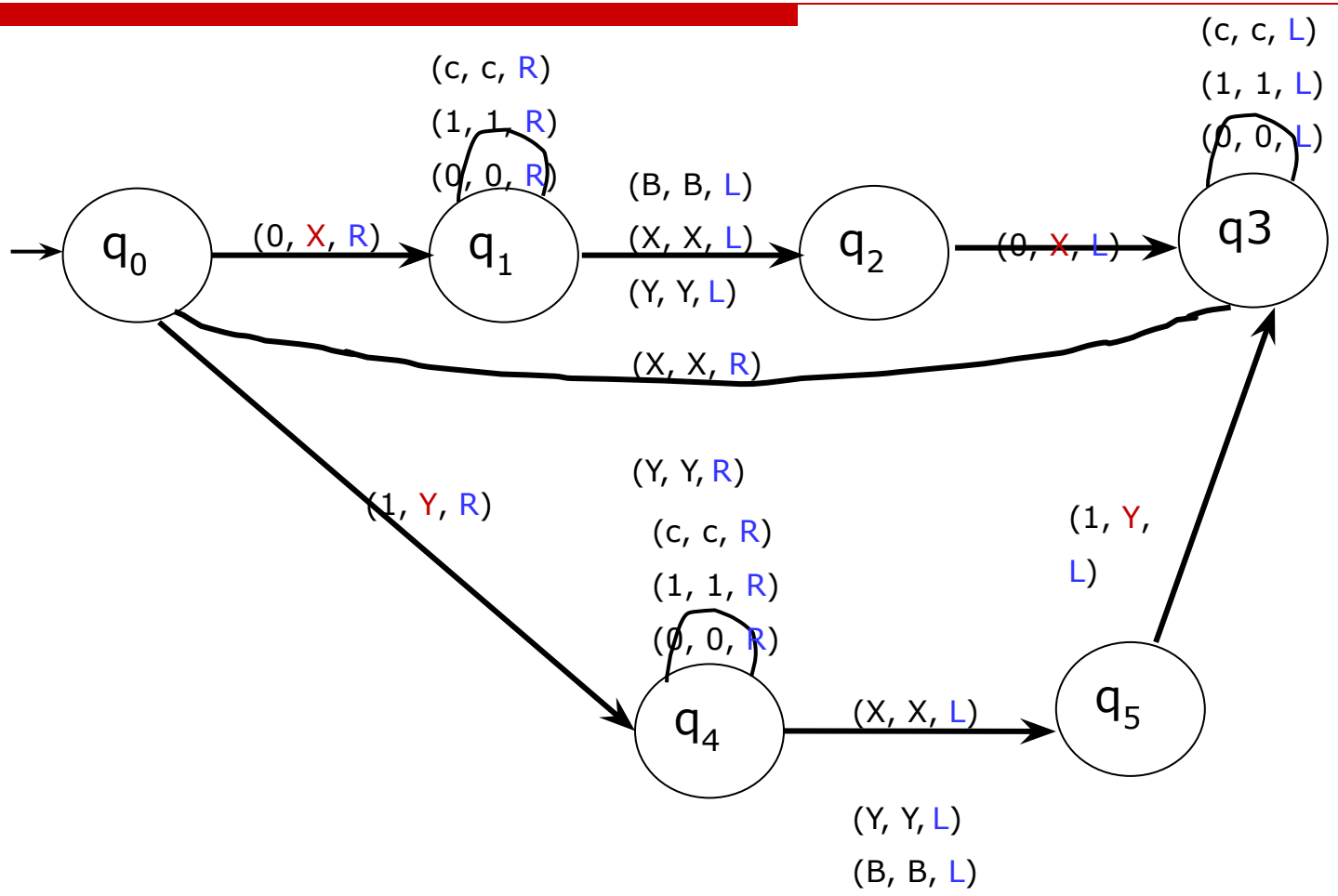
Valid string: 110c011

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | c | 0 | 1 | 1 | B |

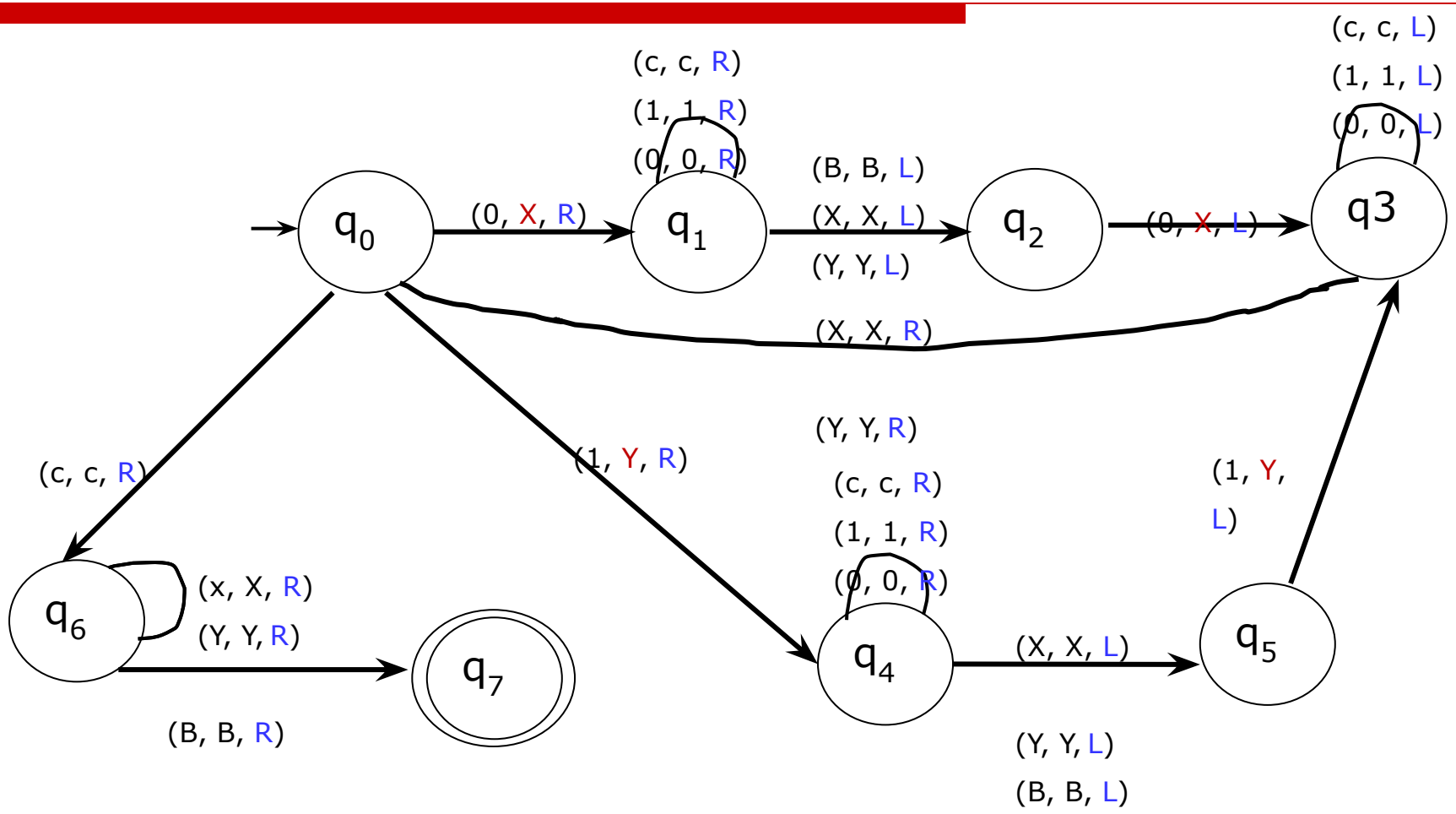
# TM for L={wcw$^r$, w ∈ (0+1)*}



# TM for L={wcw$^r$, w ∈ (0+1)*}



# Complete TM for L={wcw$^r$, w ∈ (0+1)*}



# Question

Design TM which accepts the set of all palindromes over {0, 1}

Example:

Valid Strings:

010

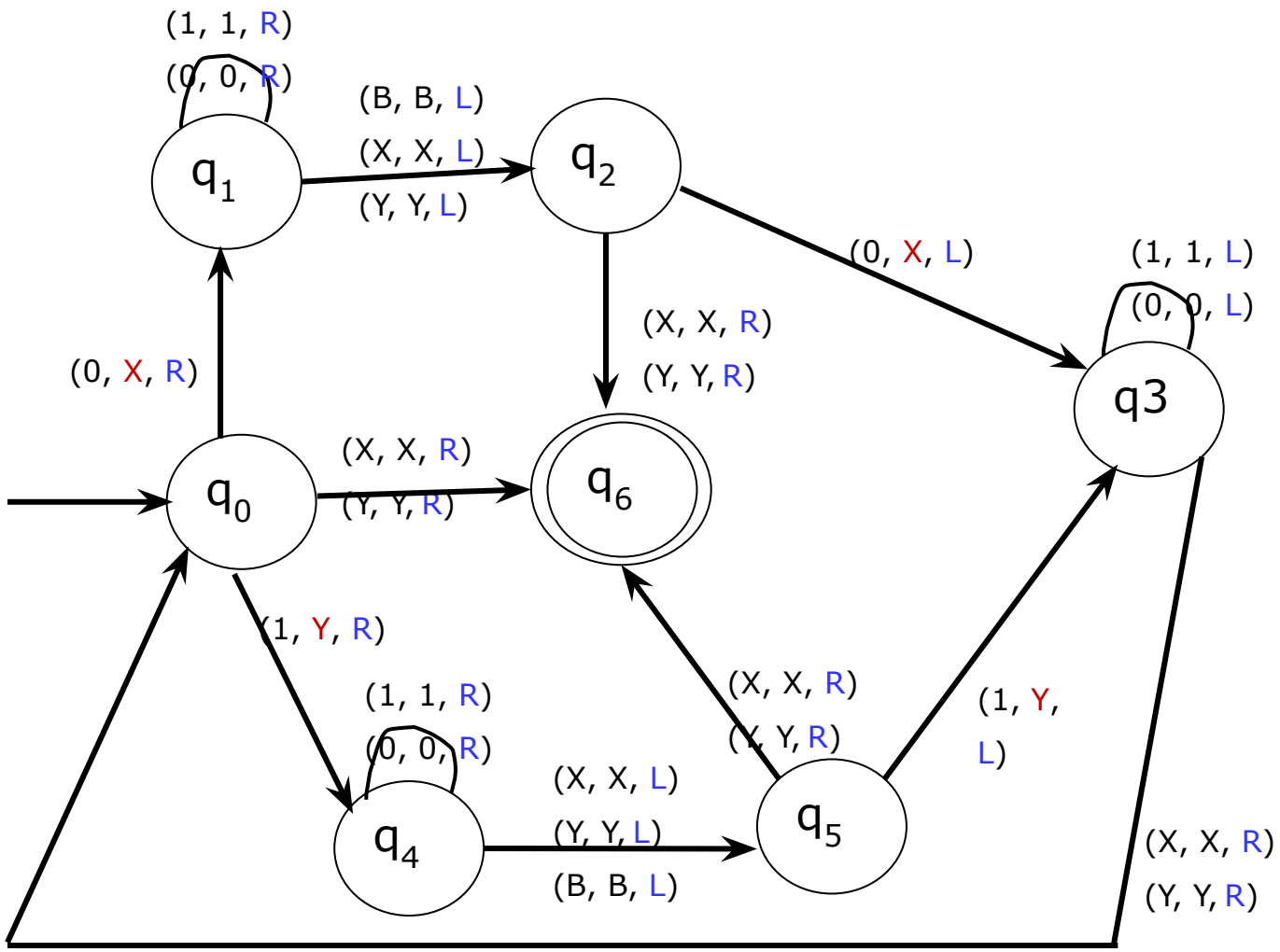
0110

Invalid Strings:

1010

110

## Complete design of TM which accepts the set of all palindromes over {0, 1}



# Question

Design Turing Machine to compute the function, which is called monus or proper subtraction and is defined by

(m-n) = max (m-n, 0)

Note:

The monus operation is defined as

(m-n) = (m-n) if m **≥** n

(m-n) = 0 if m **<** n

Example:

If m = 5 , n =2 **then** (m-n) = (5-2) = **3**

If m = 2 , n =5 **then** (m-n) = (2-5) = **0**

## Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

To start with the tape consists of $0^m10^n$ which is terminated by blank symbol.

Example:

m=5 , n=2

Input tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | B |

Output Tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | 0 | 0 | 0 | B | B | B | B |

(5-2)=3

m=2 , n=5

Input tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | B |

Output Tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | B | B | B | B | B | B | B |

(2-5)=0

Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

Rough Slide

To start with the tape consists of $0^m10^n$ which is terminated by blank symbol.

Example:

m=2 , n=5

Input tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | B |

Output Tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | B | B | B | B | B | B | B |

(2-5)=0

Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

Rough Slide

To start with the tape consists of $0^m10^n$ which is terminated by blank symbol.

Example:

m=5 , n=2

Input tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | B |

Output Tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | 0 | 0 | 0 | B | B | B | B |

(5-2)=3

## Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

To start with the tape consists of $0^m10^n$ which is terminated by blank symbol.

Example: m=5 , n=2

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | B |

**General Approach:** The sequence of 0's is partitioned into first group with **m** number of 0's followed by a 1 and followed by second group with **n** number of 0's. The machine finds leftmost 0 and is replaced by blank B. then move towards right to search for 1. After finding 1, it search's leftmost 0 in the second group and is replaced by 1 and move towards left to get leftmost 0 in the first group. The procedure is repeated till one of the following conditions are satisfied:

- When searching for a 0 in second group, if B is encountered it means that **n** number of 0's in the second group are replaced by 1's and (n+1) in the first group are changed to B's. Now, the second group will have (n+1) ones. The machine replaces (n+1) 1's by one 0 and **n** B's and observe that only (m-n) 0's exists on the tape.

- In the first group, if the machine can not find a 0 (Since first m 0's have already been changed to B's) it mean that (m<n) and so no 0's and 1's should be there on the tape.

# Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

Input tape

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | B |

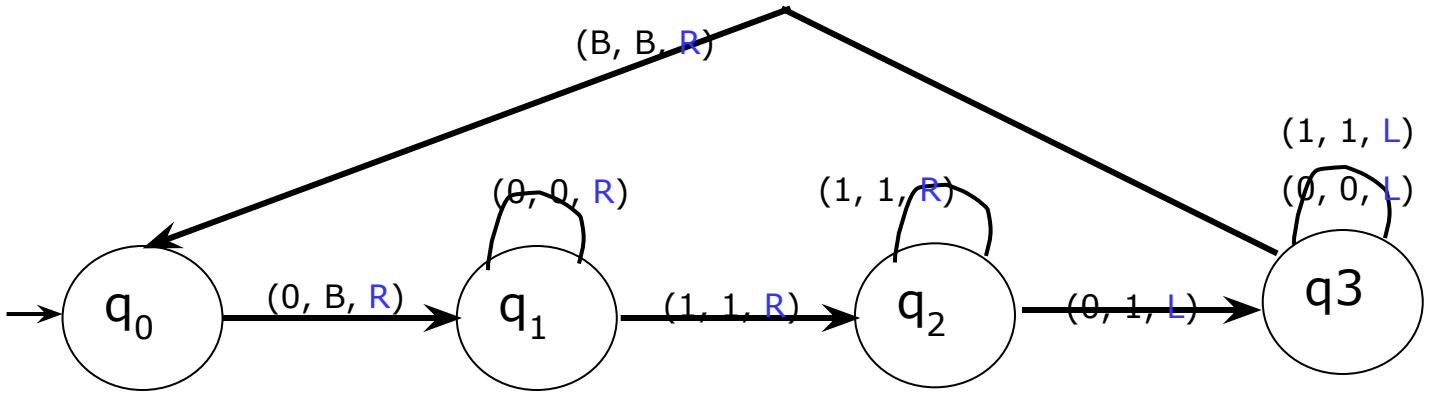
Output Tape (5-2)=3

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | B | 0 | 0 | 0 | B | B | B |

Logic

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | B |

**Step 1:**
In state $q_0$, if 0 is encountered
- Replace 0 by B, Change state to $q_1$, Move right.
- Replace 0 by 0, be in same state $q_1$, Move right. When 1 is encountered, Change State to $q_2$, Move Right.
- Replace 1 by 1, be in same state $q_2$. When leftmost 0 is encountered replace by 1, change state to $q_3$, move left. When B is encountered replace by B, Move left, change state to q4, go to step 2
-Replace 0 by 0, 1 by 1, be same state $q_3$ . When B is encountered, replace by B, move right, go to state $q_0$. Repeat step 1
In state $q_0$, if 1 is encountered, go to step 3



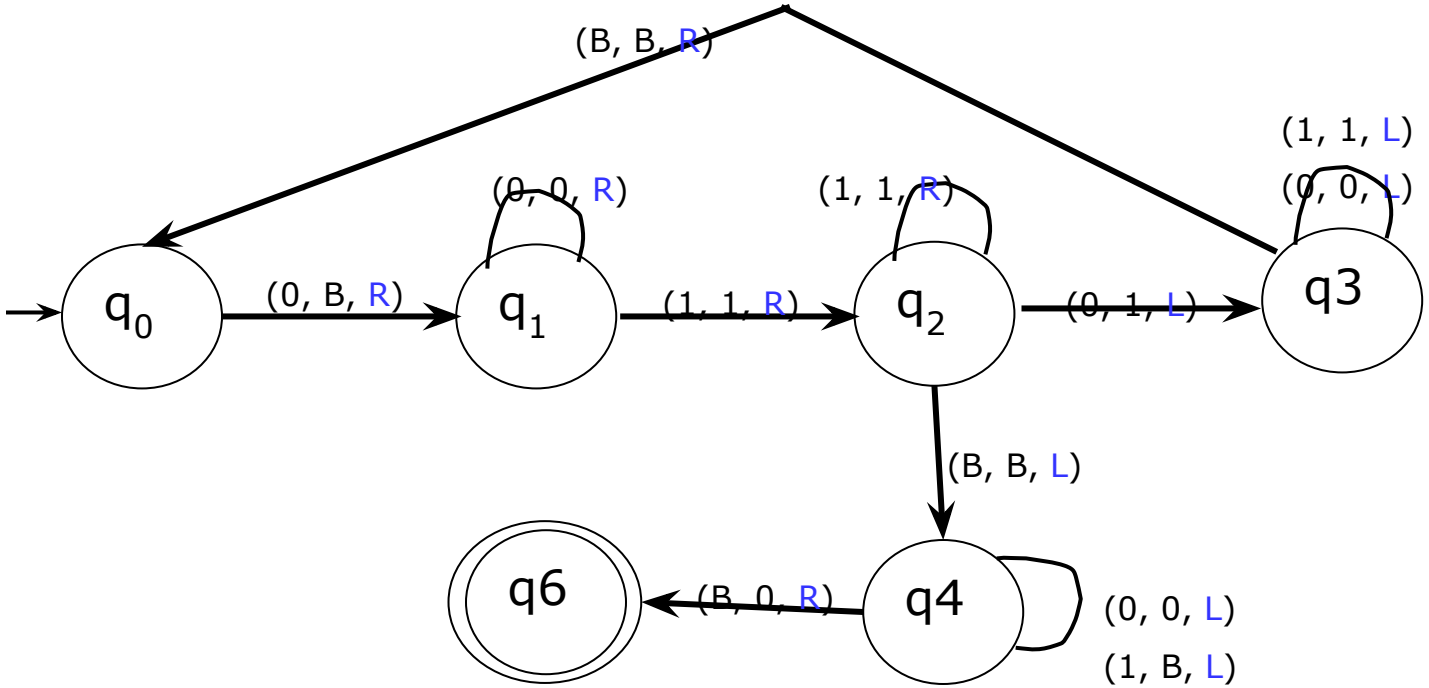
# Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

Logic

Example for (5-2)=3

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | B | 0 | 0 | 1 | 1 | 1 | B |

Step 2:
- Replace 0 by 0, Replace 1 by B, be in same state q4, Move left.
- If B is encountered replace by 0, move right, go to final state $q_6$



# Design TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)

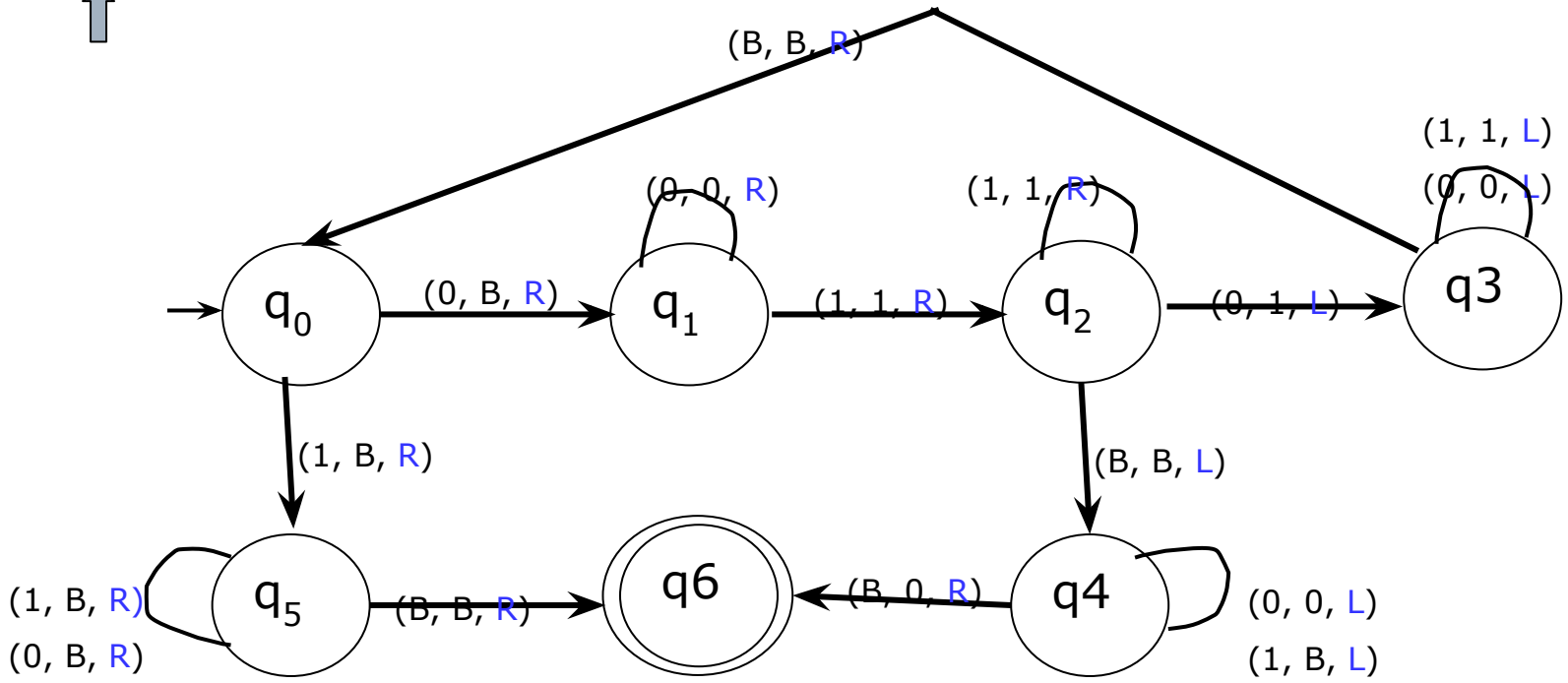
Input

Example for (2-5)=0

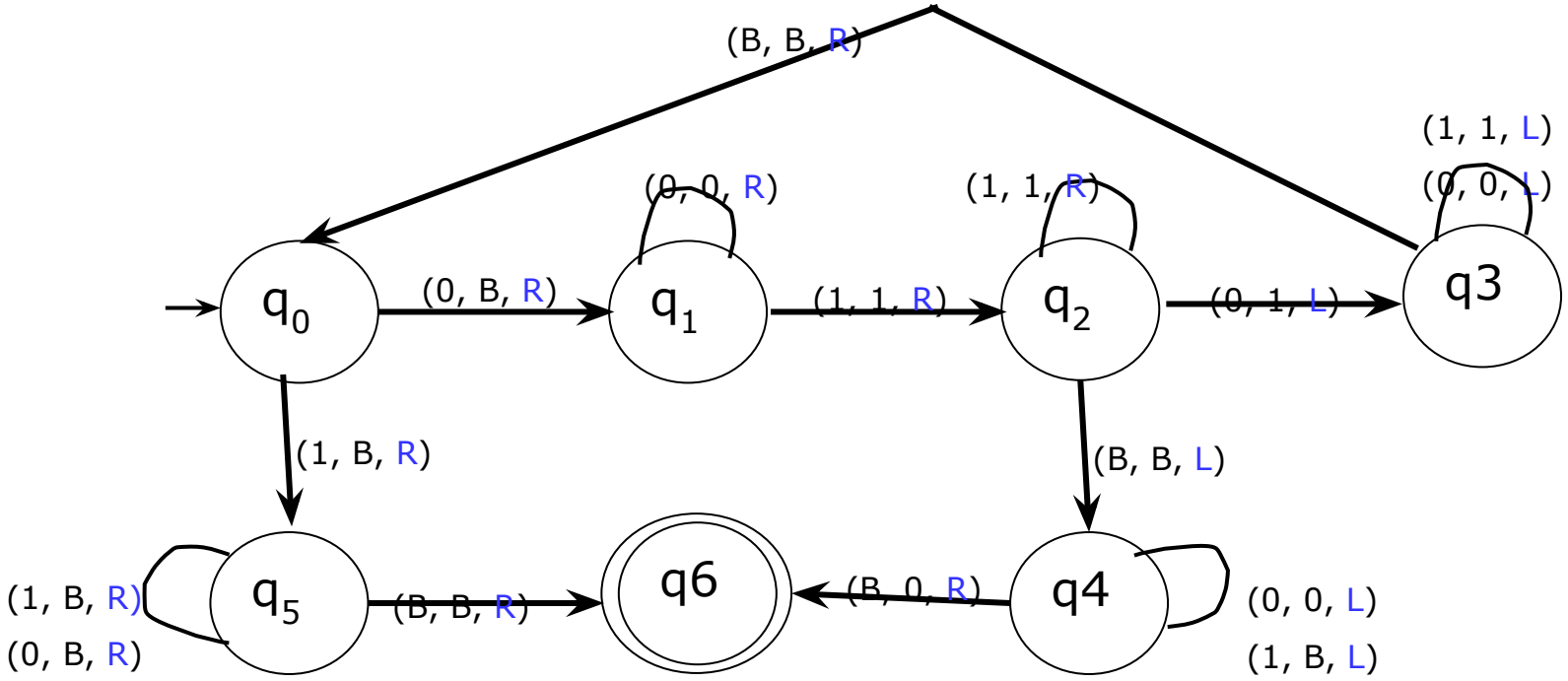
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | B |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B | B | 1 | 1 | 1 | 0 | 0 | 0 | B |

Step 3:
- Replace 1 by B, Replace 0 by B, be in same state q5, Move right.
- If B is encountered replace by B, move e right, go to final state $q_6$



Complete design of TM to compute the function, which is called monus or proper subtraction and is defined by (m-n) = max (m-n, 0)



# Multiplication in TM (using Subroutines)

Example:

2x3 = 6

Input Tape:

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 | 1 | 0 | B | B | B | B | B | B | B |

Output Tape:

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | B | B | B | B | B | B | 1 | 1 | 1 | 1 | 1 | 1 | B |

# Multiplication in TM (using Subroutines)
## Rough Slide

Example: 2x3 = 6

Input Tape:

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 | 1 | 0 | B | B | B | B | B | B | B |

# Multiplication in TM (using Subroutines)

Input Tape

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 | 1 | 0 | B | B | B | B | B | B | B |

Output Tape

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | B | B | B | B | B | B | 1 | 1 | 1 | 1 | 1 | 1 | B |

# Question

Design TM for L={SS, S ∈(a+b)*)

Example:

Valid Strings:

aa

abab

aabaab

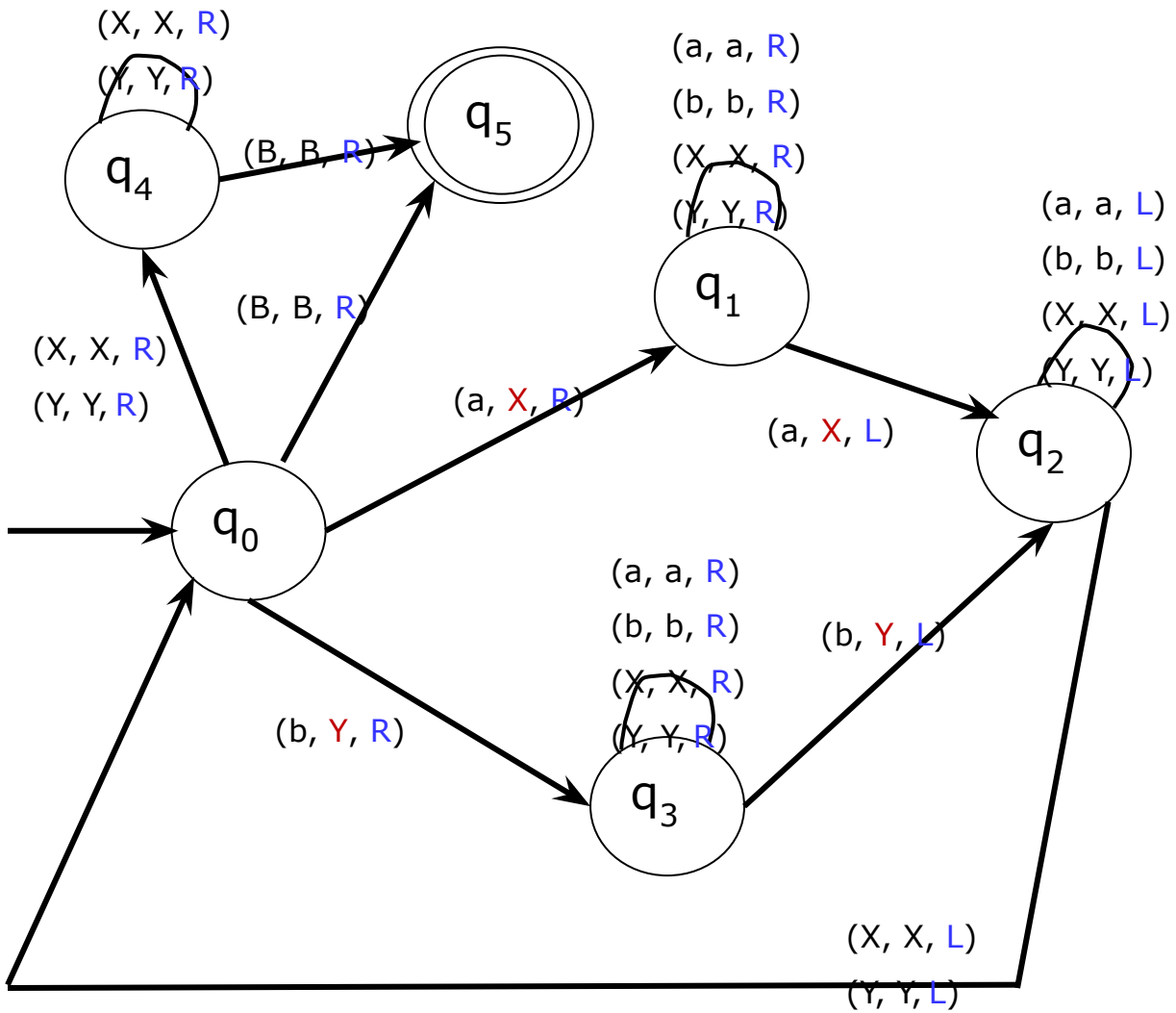
Invalid Strings:

ab

aaa

abba

# Complete TM for L={SS, S ∈(a+b)*)

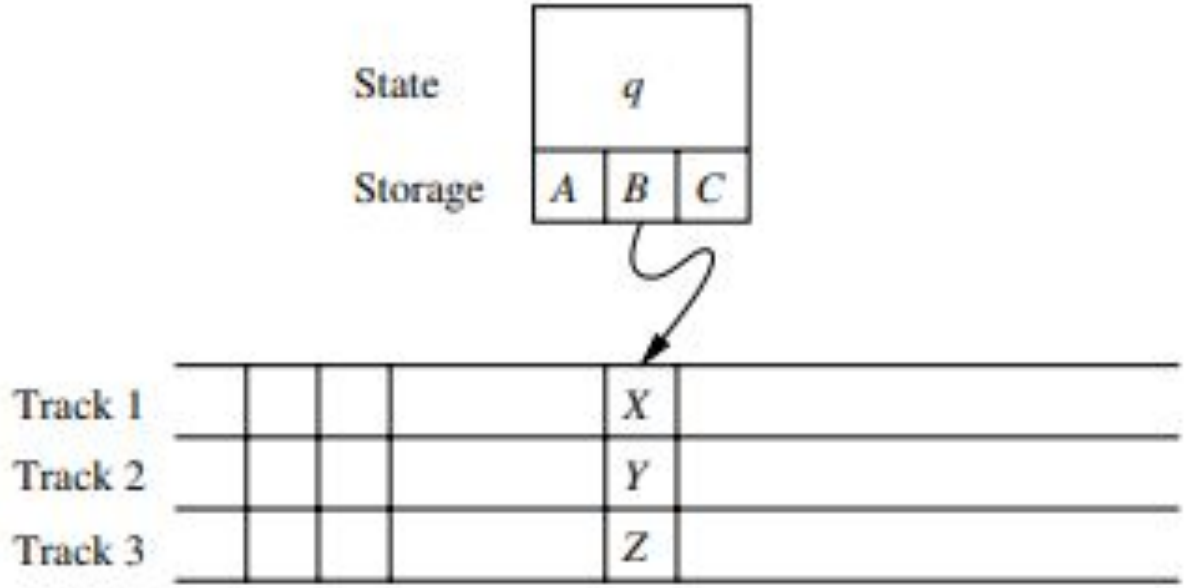


# Programming Techniques for Turing Machine (TM)

- Storage in the state
- Multipule Tracks
- Subroutines

# Storage in the State

We can use the finite control not only to represent a position in the program of the Turing machine but to hold a finite amount of data Figure



# Mutiple Tracks TM and Subroutine TM

**Mutiple Tracks TM:**

Another useful trick is to think of the tape of a Turing machine as composed of several tracks. Each track can hold one symbol and the tape alphabet of the TM consists of tuples with one component for each.

**Subroutine TM:**

As with programs in general it helps to think of Turing machines as built from a collection of interacting components or subroutines A Turing machine subroutine is a set of states that perform some useful process. This set of states includes a start state and another state that temporarily has no moves and that serves as the return state to pass control to whatever other set of states called the subroutine. The call of a subroutine occurs whenever there is a transition to its initial state. Since the TM has no mechanism for remembering a return address that is a state to go to after it finishes, should our design of a TM call for one subroutine to be called from several states we can make copies of the subroutine using a new set of states for each copy. The calls are made to the start states of different copies of the subroutine and each copy returns to a different states.

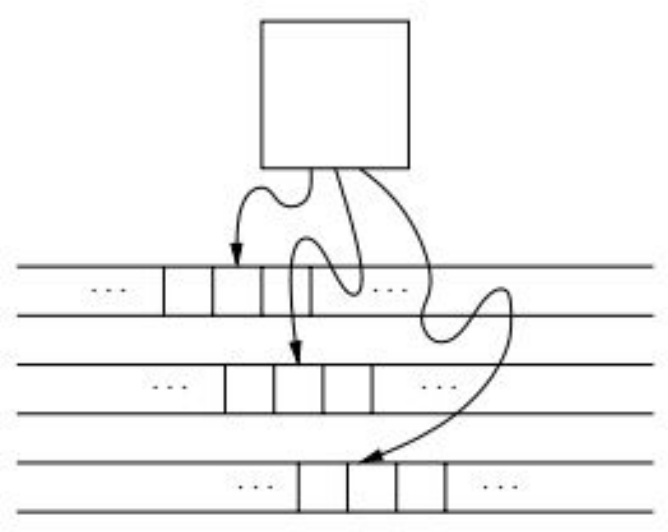
# Extension of Basic Turing Machine (TM)

- Multitape TM

# Mutitape TM

A multitape TM is as suggested by Fig below. The device has a finite control state and some finite number of tapes Each tape is divided into cells and each cell can hold any symbol of the finite tape alphabet As in the single tape TM the set of tape symbols includes a blank and has a subset called the input symbols of which the blank is not a member. The set of states includes an initial state and some accepting states Initially

1. The input a finite sequence of input symbols is placed on the first tape
2. All other cells of all the tapes hold the blank
3. The finite control is in the initial state
4. The head of the first tape is at the left end of the input
5. All other tape heads are at some arbitrary cell Since tapes other than the first tape are completely blank it does not matter where the head is placed initially all cells of these tapes look the same



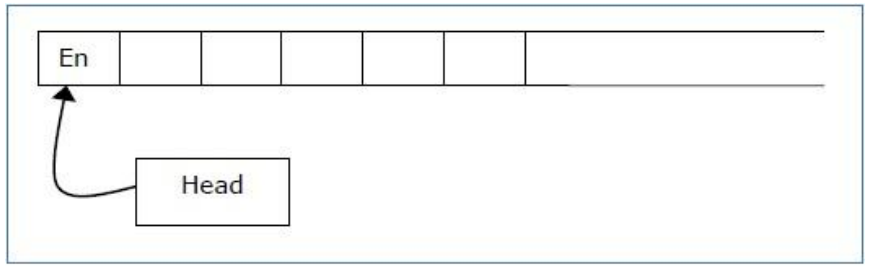
Multitape TUM

# Restricted Turing Machine (TM)

- Semi-infinite tape TM
- Multi-stack TM

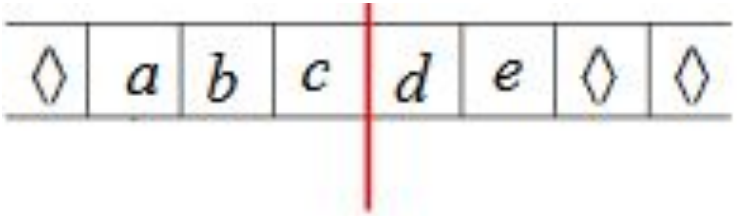
# Semi-infinite Tape machine

A Turing Machine with a semi-infinite tape has a left end but no right end. The left end is limited with an end marker.



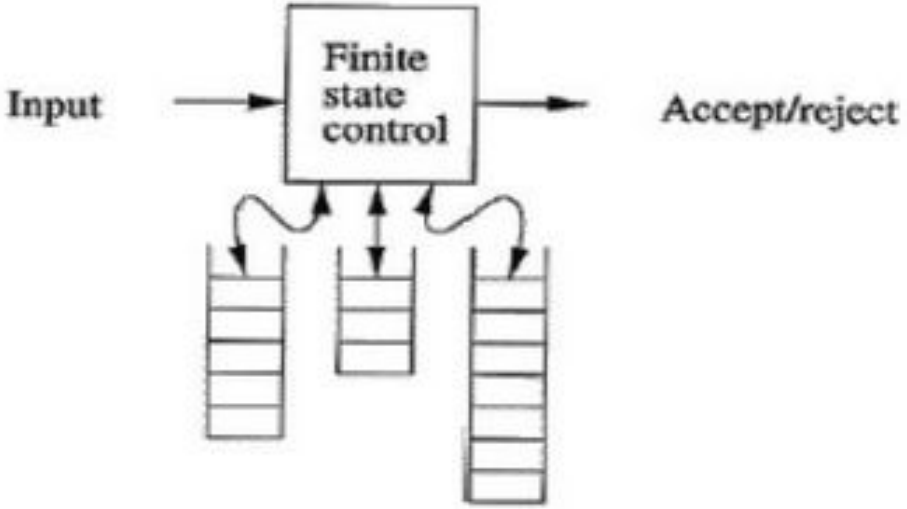

It is a two-track tape –

- **Upper track** – It represents the cells to the right of the initial head position.
- **Lower track** – It represents the cells to the left of the initial head position in reverse order.



# Multi-stack Turing Machine

- Generalizations of the PDAs
- TMs can accept languages that are not accepted by any PDA with one stack.
- But PDA with two stacks can accept any language that a TM can accept.



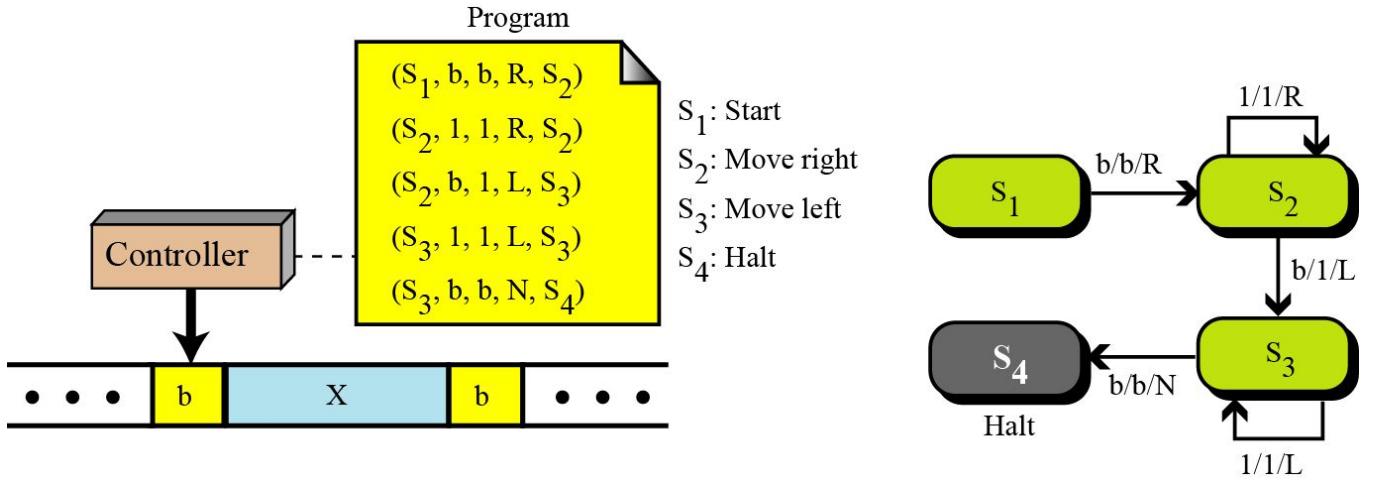
# Multi-stack Turing Machine

- A k-stack machine is a deterministic PDA with k stacks.
- It obtains its input from an input source rather than having the input placed in a tape.
- It has a finite control, which is in one of a finite set of states.
- It has a finite stack alphabet, which it uses for all its stacks.
- A move of a multistack machine is based on:
  - The state of the finite control
  - The input symbol read, which is chosen from the finite input alphabet
  - The top stack symbol on each stack

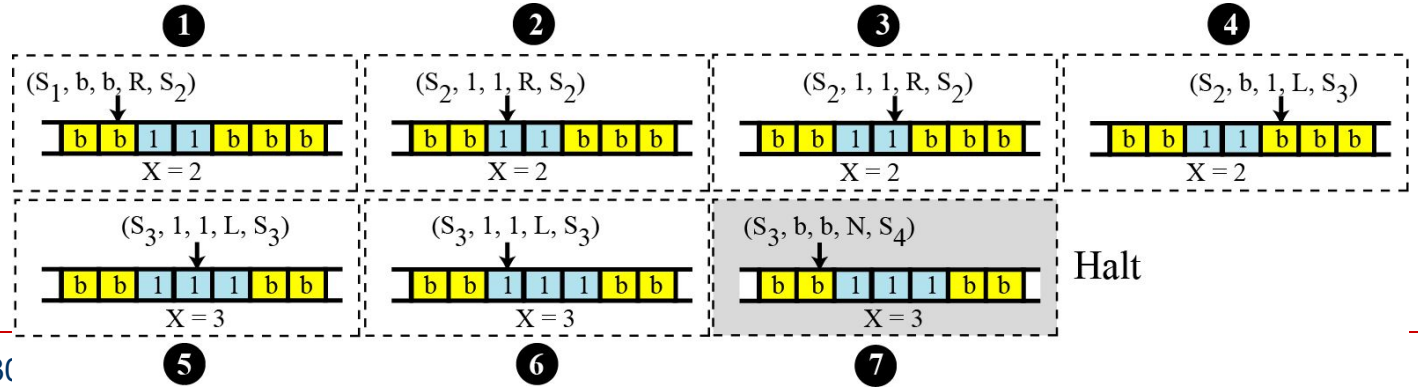
# Turing Machine and Computers

# Simulating computer by Turing machine

- Example (1): Increment Operation[ incr(x) ]

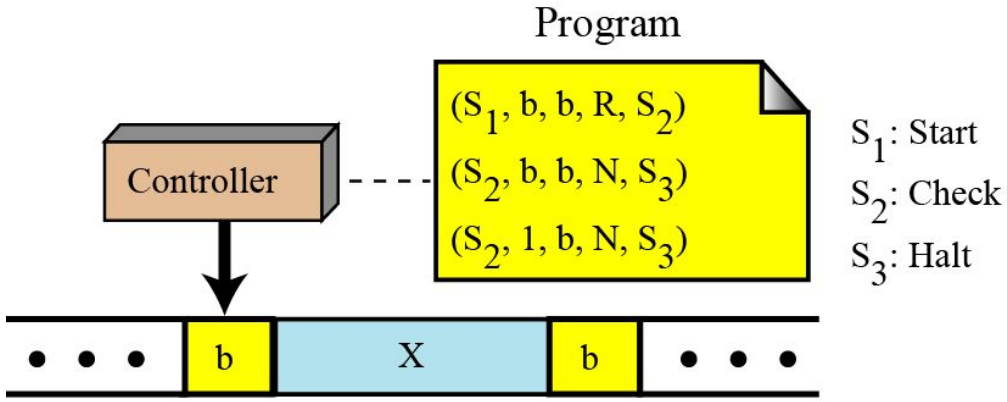


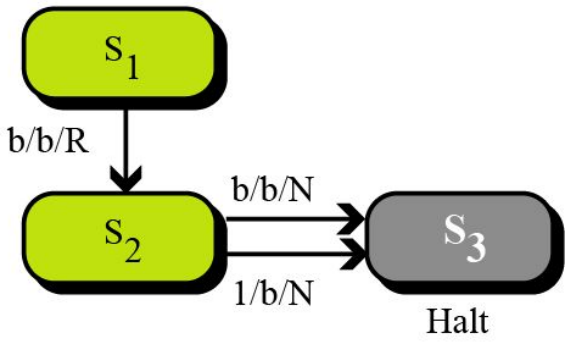

- Step by step operation which shows how the TM can increment x when x=2



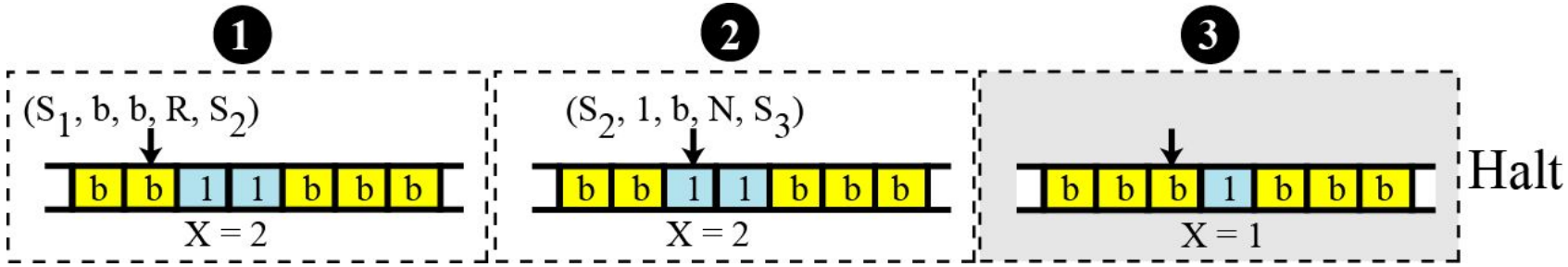
# Simulating computer by Turing machine

- Example (2): Decrement Operation[ decr(x) ]




- Step by step operation which shows how the TM can decrement x when x=2



# Introduction to Undecidable Problems

- Post Correspondence problem

# Recursive and Recursively Enumerable language

**Recursive Language:**

- A language 'L' is said to be recursive if there exists a Turing machine which will accept all the strings in 'L' and reject all the strings not in 'L'.
- The Turing machine will halt every time and give an answer (accepted or rejected) for each and every string input.

**Recursively Enumerable Language:**

- A language 'L' is said to be a recursively enumerable language if there exists a Turing machine which will accept (and therefore halt) for all the input strings which are in 'L'.
- But may or may not halt for all input strings which are not in 'L'.

# Decidable, Partially-Decidable and Undecidable language

**Decidable Language:**

A language 'L' is decidable if it is a recursive language. All decidable languages are recursive languages and vice-versa.

**Partially Decidable Language:**

A language 'L' is partially decidable if 'L' is a recursively enumerable language.

**Undecidable Language:**

- A language is undecidable if it is not decidable.
- An undecidable language may sometimes be partially decidable but not decidable.
- If a language is not even partially decidable, then there exists no Turing machine for that language

# Undecidable Problems

| | |
|---|---|
| Recursive Language | TM will always Halt |
| Recursively Enumerable Language | TM will halt sometimes & may not halt sometimes |
| Decidable Language | Recursive Language |
| Partially Decidable Language | Recursively Enumerable Language |
| UNDECIDABLE | No TM for that language |

# Undecidable problems

- Some problems take a very long time to solve, so we use algorithms that give approximate solutions. There are some problems that a computer can *never* solve, even the world's most powerful computer with infinite time.
- An **undecidable problem** is one that should give a "yes" or "no" answer, but yet no algorithm exists that can answer correctly on all inputs.

# Halting Problem

Alan Turing proved the existence of undecidable problems in 1936 by finding an example,
the now famous "halting problem":
*Based on its code and an input, will a particular program ever finish running?*
For example, consider this program that counts down:

```
num ← 10
REPEAT UNTIL (num = 0) {
        DISPLAY(num)
        num ← num – 1
 }
```

That program will halt, since num eventually becomes 0.
Compare that to this program that counts up:

```
num ← 1
REPEAT UNTIL (num = 0) {
        DISPLAY(num)
        num ← num + 1
}
```

It counts up *forever*, since num will never equal 0.

Algorithms *do* exist that can correctly predict that the first program halts and the second program never does.
These are simple programs which don't change based on different inputs.

However, no algorithm exists that can analyze *any* program's code and determine whether it halts or not.

# Post Correspondence Problem

- The Post Correspondence Problem (PCP), introduced by Emil Post in 1946, is an undecidable decision problem. The PCP problem over an alphabet ∑ is stated as follows –
- Given the following two lists, **M** and **N** of non-empty strings over ∑ –

  $M = (x_1, x_2, x_3, \ldots\ldots\ldots, x_n)$

  $N = (y_1, y_2, y_3, \ldots\ldots\ldots, y_n)$
- We can say that there is a Post Correspondence Solution, if for some $i_1, i_2, \ldots\ldots\ldots\ldots i_k$, where $1 \leq i_j \leq n$, the condition $x_{i1} \ldots\ldots x_{ik} = y_{i1} \ldots\ldots y_{ik}$ satisfies.

# Post Correspondence Problem

**Example 1:** Find whether the lists
M = (abb, aa, aaa) and
N = (bba, aaa, aa) have a Post Correspondence Solution?

**Solution**

| | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|
| **M** | Abb | aa | aaa |
| **N** | Bba | aaa | aa |

Here, $\mathbf{x_2x_1x_3}$ **= 'aaabbaaa'** and $\mathbf{y_2y_1y_3}$ **= 'aaabbaaa'**
We can see that $\mathbf{x_2x_1x_3 = y_2y_1y_3}$
Hence, the solution is **i = 2, j = 1, and k = 3**

# Post Correspondence Problem

**Example 2:** Find whether the lists **M = (ab, bab, bbaaa)** and **N = (a, ba, bab)** have a Post Correspondence Solution ?

Solution:

| | $x_1$ | $x_2$ | $x_3$ |
|---|---|---|---|
| **M** | ab | bab | bbaaa |
| **N** | a | ba | bab |

In this case, there is no solution because –

$| x_2x_1x_3 | \neq | y_2y_1y_3 |$ (Lengths are not same)

# Post Correspondence Problem

## Example

| | A | B |
|---|---|---|
| i | $w_i$ | $x_i$ |
| 1 | 1 | 111 |
| 2 | 10111 | 10 |
| 3 | 10 | 0 |

This PCP instance has a solution: 2, 1, 1, 3:

$$w_2w_1w_1w_3 = x_2x_1x_1x_3 = 101111110$$

# Post Correspondence Problem (PCP)

Does this PCP instance have a solution?

| | A | B |
|---|---|---|
| i | $w_i$ | $x_i$ |
| 1 | 110 | 110110 |
| 2 | 0011 | 00 |
| 3 | 0110 | 110 |

This PCP instance has a solution: 2,3,1

$w_2w_3w_1 = x_2x_3x_1 = 00110110110$

One more solution:

2,1,1,3,2,1,1,3

# Thanks for Listening

# Program Halting Problem

- **Input:** a program $P$ in some programming language
- **Output: true** if $P$ terminates; **false** if $P$ runs forever.

# Examples

halts("2+2")

**True**

halts("**def f(n):**
**if n==0: return 1**
**else: return n * f(n-1)**
**f(10)**")

")

**True**

halts("**def f(n):**
**if n==0: return 1**
**else: return n * f(n-1)**
**f(10.5)**")

**False**

# Tougher Example

```
halts("
    def isPerfectNumber(n): # n is perfect if factors sum to n
    divs = findDivisors(n)
    return n == sum(divs)
     i = 3
     while not isPerfectNumber (i): i = i + 2
     print i")
```

**Unknown**

Note: it is unknown where an odd perfect number exists. (Numbers up to $10^{300}$ have been tried without finding one yet.)